

IDENTIFYING DEEPPFAKE IMAGES

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FACULTY OF INFORMATION SCIENCE & TECHNOLOGY

MULTIMEDIA UNIVERSITY

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DECLARATION

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ABSTRACT

Deepfake image created by modern generative models are increasingly threatening digital trust, cybersecurity, and forensic analysis. In this project, a hybrid ensemble based deepfake image detection system that integrates RGB based deep learning models with artifact focused forensic cues is proposed to enhance detection reliability and interpretability. The system integrates ResNet50, Xception, and Vision Transformer with frequency domain analysis, Discrete Fourier Transform (DFT) and structural analysis, Canny edge detection to identify images. Afterwards, a majority voting strategy with confidence based tie breaking is used to make the final decision. Two publicly available Kaggle datasets, a StyleGAN2 based dataset containing 13,000 images and a large scale OpenForensics dataset having close to 190,000 images were used on this experiments, and evaluation of performance was done through accuracy, precision, recall and F1 score. The outcomes point that proposed in this paper, ensemble detector, has achieved steady and highly efficient performance on both datasets, has surpassed the individual artifact based detectors in term of accuracy, reliability and its still remains very strong in precision and recall. An interactive testing and forensic evidence visualization tool has also been developed on a Flask based interface and demonstrated that multi cue fusion offers a stronger and more straightforward methodology for deepfake image detection.

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LIST OF ABBREVIATIONS/ SYMBOLS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DFT	Discrete Fourier Transform
EIS	Edge Irregularity Score
GAN	Generative Adversarial Network
HFE	High-Frequency Energy
RGB	Red Green Blue
SBI	Self-Blended Image
ViT	Vision Transformer

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CHAPTER 1

INTRODUCTION

1.1 Overview

Deepfakes are variants of synthetic media that are artificially generated by advanced AI algorithms, which are capable of manipulating or fabricating images and videos to such an extent that they are visually indistinguishable from genuine ones. Due to the human nature of potential misuse in areas like cybersecurity, social media, and journalism, deepfakes have become a hot topic of research and the public concerns (Amerini et al., 2025; Liu et al., 2024). The advent of powerful generative models like GANs and the more recent diffusion-based models has greatly enhanced the quality of the generated media, to an extent that has become quite a challenge even for the human eye to verify the difference between manipulated and genuine content (Croitoru et al., 2024; Amerini et al., 2025). These abilities not only facilitate the dissemination of false information but also the identity theft, hence resulting in a decrease of trust in online information sources.

As a result, a large portion of the research community efforts has been directed towards automatic recognition of deepfake videos through machine learning and forensic analysis methods (Amerini et al., 2025; Khan et al., 2025). The methods of detection vary mostly from traditional image forensic features to deep learning-based classifiers, with the focus being on avoiding overfitting to specific generative models and datasets. For carrying out empirical studies and experiments, this paper will take the advantage of the two massive datasets sourced from Kaggle, which have become the staple datasets in academic and community research circles due to their open source and the diverse collection of real and fake images they contain and facilitating reproducible testing and comparison of deepfake detection algorithms.

1.2 Problem Statement

In an earlier time, deepfake detection methods mostly tackled the problem through visible artifacts or inconsistencies in the deepfakes such as abnormal blinking behavior. However, these methods are not reliable as those deepfake continue to improve in generation quality and the attack methods keep evolving (Li, Chang, & Lyu, 2018; Liu, Tao, & Zhou, 2025; Sunil et al., 2025). Now in a recent year, more attention is being criticize in forensic literature to the idea that we should not stop at identifying "surface errors" but also trace the forensic elimination method or model so we can fingerprint the very process of synthesis and attribute the forgery (Amerini et al., 2025; Bayar & Stamm, 2018). So, the work of such forensic systems capable of associating deepfakes with their underlying generative mechanisms beyond just detecting the visible artifacts becomes a very critical milestone (Amerini et al., 2025; Liu, Tao, & Zhou, 2025).

There are two main limitations of strategies still extensively present in many current deepfake recognition approaches. To begin with first, black box deep models like CNNs and transformers can perform very well in performance with their trained and known dataset, but they mostly do not work well cross generator or cross domain, and they provide little interpretability, which is essential for forensic use (Amerini et al., 2025; Liu, Tao, & Zhou, 2025; Sunil et al., 2025). Secondly, single cue visual artifact methods such as boundary or blending anomalies are although can be very effective, they still may not enough in the face of rapid synthesis (Li et al., 2020; Liu, Tao, & Zhou, 2025). As a result, a more solid and proofed direction is ensemble style forensic detector that purposely focus on manipulation traces and mutually exclusive artifact families, for instance: boundary evidence coupled with signal level traces. Therefore, it can raises both robustness and explainability aspects (Amerini et al., 2025; Bayar & Stamm, 2018; Li et al., 2020).

Benchmarking practices can differ drastically in various papers and implementations, so it is very hard to tell if artifact focused methods really bring about more robustness and generalization (Amerini et al., 2025; Sunil et al., 2025). Furthermore, using accuracy as the only measure can be very deceptive especially in the case of class imbalance so reporting precision, recall and F1 might becomes crucial

in obtaining a credible evaluation as well as for a fair comparison between different datasets and operating points (Chicco & Jurman, 2020; Saito & Rehmsmeier, 2015). If benchmarking is more comprehensive and standardized, it would be better on validation claims regarding component specific artifacts versus traditional detection cues (Amerini et al., 2025; Sunil et al., 2025).

1.3 Project Objectives

Problem Statement	Corresponding Objective
1	Through study of relevant literature and initial experiments of Kaggle datasets, then identify the artifacts from samplers that have been produced in deepfake images, using ensemble hybrid-detection models.
2	Use python to create a detection tool that employs OpenCV and PyTorch to build up a ensemble detector that integrate 3 main RGB architectures, ResNet50, Xception, and Vision Transformer while also cooperated traditional artifact-focused analysis DFT, Canny Edge Detection with voting-based fusion. Make a Flask web interface for interactive detection and forensic evidence visualization.
3	Measure the ensemble effectiveness on the two Kaggle datasets using accuracy, precision, recall, and F1-score. Benchmark performance against baseline methods like XceptionNet, edge-only detectors and visualize results via confusion matrices, confidence distributions, and result counts.

Table 1.1: Problem statements and objective

1.4 Project Scope

This work is limited to the detection of image-based deepfake only, and the focus on videos content will be convert to video frames for detection. This research utilizes two large-scale datasets sourced from Kaggle to ensure high fidelity and diversity. The first dataset consists of a balanced set of 13,000 images (Real vs. Fake), and the second dataset is a comprehensive 190,000 images. All datasets will fully train on Google Colab. The detection process will be carried out through the identification of blending inconsistencies, texture loss, or unnatural frequency patterns, among other things that can be measured in fake images.

CHAPTER 2

LITERATURE REVIEW

Deepfake technology has changed drastically over the last ten years which is mainly due to the generative models and computer vision improvements. As a result, detection tools are now in a constant battle to catch up, but they still have significant weaknesses in understanding, generalizing, and pinpointing targeted artifacts. This literature discusses the deepfake generation evolution, getting deep into detection issues and the limitations that led in the current research.

2.1 Deepfake Generation and Artifact Formation

Deepfake images are a artificial image that has been created by deep generative models. These methods can generate realistic content artificially when the intricacy of visual patterns is learned. It is becoming less and less easy to visually distinguish deepfakes as the production quality improves. Studies have shown most advanced deepfakes are no longer rely on visible errors but carry extremely subtle model specific fingerprints which can be uncovered through statistical and signal level analysis (Ricker et al., 2024; Aghasanli et al., 2023). Such model fingerprints exist in both spatial and frequency domain, which is why multi domain detection methods are necessary. Figure 1 shown are examples of frequency domain representations of genuine images and those different GAN and diffusion model generators:

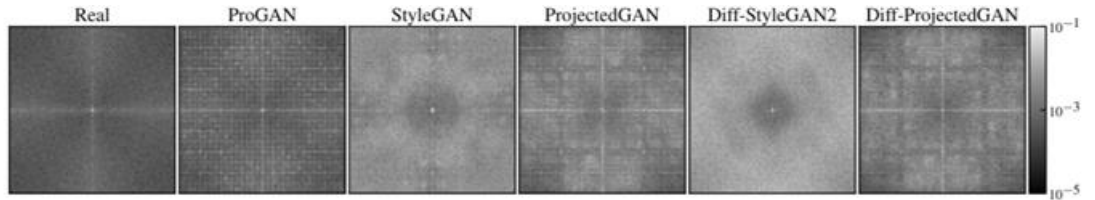


Figure 2.1: Frequency domain comparison of real and generated images (adapted from Ricker et al., 2024).

The figure shows that every generative model is characterized with their own characteristic spectral distribution which implies frequency based features can be point to generator specific fingerprints. The first deepfakes to appear were mainly GAN

based on lead to artifacts typical of texture synthesis, upsampling, and blending. Newer versions of GANs possibly have fewer visible defects however detectable texture and frequency anomalies still works (Hasanaath et al., 2024). Diffusion models are one step ahead of GANs in terms of realism; however, they present a different spectral pattern such as suppression of high frequency components (Ricker et al., 2024). In addition to that, blending and alignment processes can also cause small irregularities at the edges (Shiohara & Yamasaki, 2022). To sum up, it is right to say deception through deepfakes today is not limited to a single type of clue, but rather encompasses multiple types of clues spread across different domains which in turn, makes more beneficial to employ hybrid detection techniques.

2.2 Detection Approaches

Various deepfake detection methods fall into the category of RGB-based deep learning classifiers, generalization oriented training strategies, and hybrid or ensemble systems. CNN based models stay prevalent as they are known for their strong feature extracting capabilities, however in the latest research indicates that, a significant decline in model performance during testing on generators that the model has not train before or in cross dataset scenarios (Brodaric, 2024). Besides this, Vision Transformers have been tested as well for their ability to capture the global relationships. But similarly, they suffer from the generalization problem when the domain is shifted (Yan et al., 2023).

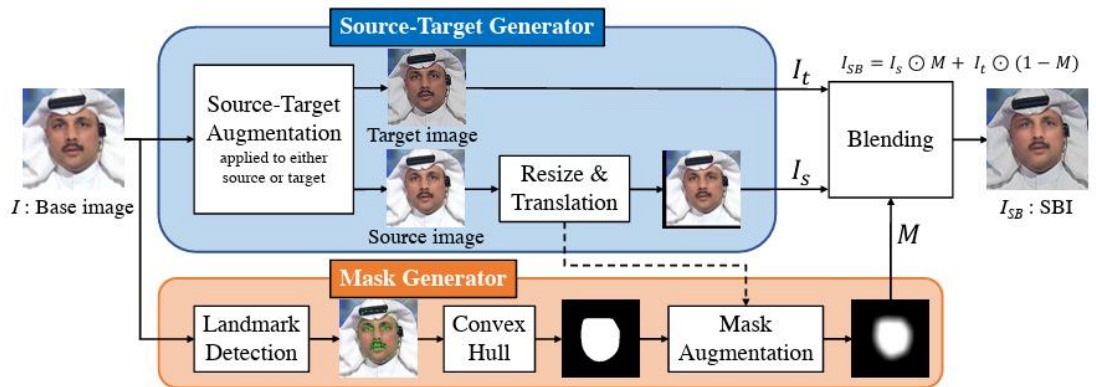


Figure 2.2: Overview of generating an SBI (Source: Shiohara & Yamasaki, 2022)

Figure 2.2 showed is one of the main strategies that used self blended image (SBI) training which imitates blending artifacts to increase robustness (Shiohara & Yamasaki, 2022). Besides all this, enhanced frequency blending also helps generalization in a more effective way (Hasanaath et al., 2024). Hybrid and ensemble detectors mix CNNs, transformers, and frequency can aware those features, which can used to identify complementary flaws and thus improve the stability across datasets (Yan et al., 2023). This is one of the advantage of ensemble based designs.

2.3 Artifact Detect Through Frequency and Structural Cues

In recent years, most of the studies in the field of detection are concentrated on artifact-aware methods. This is because contemporary deepfakes hardly show any visible flaw due to advance deepfake evolving. Nonetheless, they still have existing traces which can be exploit. These traces are mainly frequency domain distortions and structural inconsistencies (Hasanaath et al., 2024; Ricker et al., 2024), can be analysis through DFT magnitude. Hence, performing a frequency analysis is a good method since generative pipelines change the natural spectral distributions; diffusion generated images are an example of this, and in such cases, the high-frequency components might even be intentionally suppressed (Ricker et al., 2024).

There still have some artifacts remnants associated with blending, alignment and region replacement that is remain visible even if it is very subtle. Moreover, the addition of edge-lines representations can also help in reveal boundary inconsistencies and unnatural transitions, even when they are hidden in images (Shiohara & Yamasaki, 2022). The latest research demonstrates an integration of frequency and structural features leads which can enhanced robustness since it reflect different facets of the manipulation (Yan et al., 2023; Hasanaath et al., 2024). This finding is in line with hybrid approaches that combine the two domains.

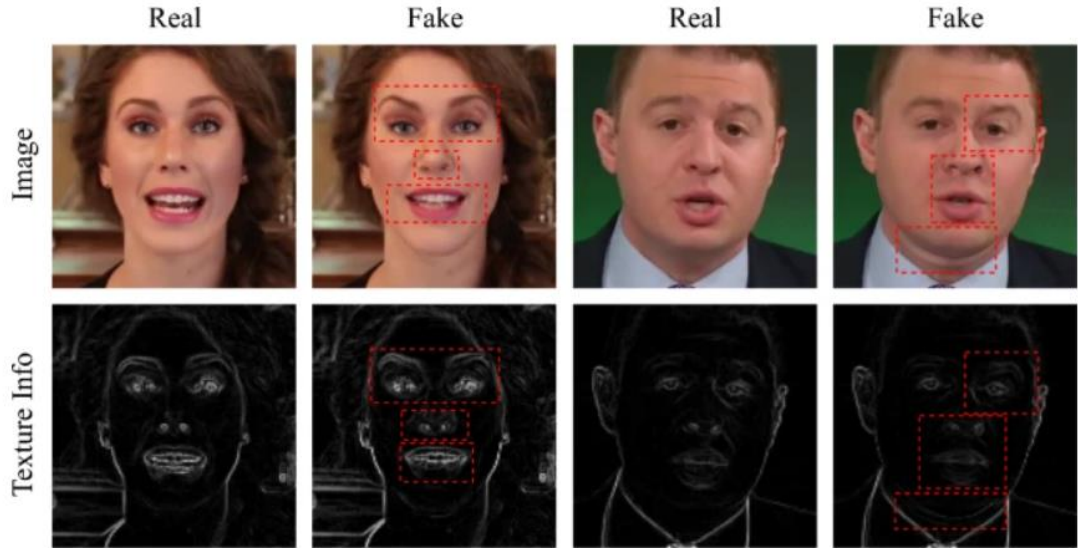


Figure 2.3: Comparison of real and deepfake images using Canny edge detection (Wen et al., 2024).

2.4 Evaluation Practices in Deepfake Detection

Yan et al. (2023) has pinpoint the lack of standardized and unified benchmark as one of the main issue in deepfake detection. They had explain the different and incompatible data processing pipelines, experimental settings, and evaluation strategies can make performance comparison seem unfair and be misunderstood. To solve this problem, they propose DeepfakeBench framework, a thorough benchmarking framework that offers unified data management, combined implementations of leading detectors as well as the standardized evaluation protocols and metrics to bring about greater transparency and reproducibility.

DeepfakeBench judges detectors using various complementary metrics. For instance, accuracy, AUC, average precision, equal error rate, precision and recall, rather than single score evaluation (Yan et al., 2023). Additionally, the authors also suggest using significance of both within-domain and cross-domain evaluation, which that can indicating majority of detectors display a considerable drop in performance when tested on unseen manipulations, thus and for all, pointing generalization as a major issue. Likewise, Brodaric (2024) had illustrates that cross dataset testing is the only way to evaluate real world robustness where the data distribution is not expected to be the same as the training set. In line with this, the present work adopts the use of

standardized evaluation procedures, gives reports on multiple metrics, and includes cross dataset testing.

2.5 Summary

Each year, deepfake detection literature is making very significant advancements. One can see this through the fact that generative model keeps making deepfakes more and more realistic, while the detection systems keep getting redesigned to counter them. Traditionally, RGB deep classifiers like CNNs and transformers tend to have strong capacity on representing features, but still have difficulty in generalization when meet with unseen generators or use cross-dataset conditions (Brodaric, 2024; Yan et al., 2023). To overcome these drawbacks, most up to date research is in favor of the use of hybrid and ensemble approaches which combine different kinds of representations for example, combination of CNNs with spatial, frequency, and structural into one detector (Hasanaath et al., 2024). The word, artifact aware analysis in this context is goes beyond simple enriching representations by learned capabilities. It is done by segmenting spectral and boundary anomalies, which is the generative processes inevitably leave behind even the high quality synthetic images are included (Ricker et al., 2024; Shiohara & Yamasaki, 2022). Likewise, latest benchmark frameworks are not only provide multiple complementary metrics, but also implement cross-domain evaluation protocols to give more thoughtful and easily understandable assessment of detector performance (Yan et al., 2023). To sum it up, these research trends substantiate the core concept of proposed hybrid ensemble detector and they enabled a very solid theoretical base for it creation and assessment in this project.

CHAPTER 3

Methodology

3.1 Research Design Overview

The methodology follows a four-stage pipeline. Each stage directly maps to the research objectives:

- **Data Preparation:** Collect and preprocess two Kaggle datasets, StyleGAN2 dataset and a OpenForensics dataset (Le et al., 2021) to support artifact analysis across real and deepfake images (supports Objective 1).
- **Component-Specific Artifact Extraction:** Employ DFT (Frequency Domain) and Canny Edge Detection (spatial domain) to isolate the synthetic artifacts (supports Objective 1 and 2).
- **Ensemble Model & Interface Development:** Build a multi-model ensemble (ResNet50, Xception, ViT, DFT CNN, Canny CNN) with voting-based fusion. Implement a Flask web interface for interactive detection (supports Objective 2).
- **Experiment & Evaluation:** Test the ensemble models performance on standard datasets using four primary metrics, benchmark against baselines as well as visualize results (supports Objective 3).

3.2 Data Preparation

3.2.1 Dataset Selection

To build and train a hybrid-model ensemble with ResNet50, Xception, ViT, DFT, and Canny that were mentioned in objective 2 and 3 as well as compatible with the research benchmarks, two public available Kaggle datasets were selected in this research:

1. StyleGan2 Deepfake Face Images (Kshitiz Bhargava, 2024):
 - Possessed 13,000 high-resolution facial images (7000 real; 6000 fake).
 - Dataset has splits into real and fake categories. This serves as the primary dataset for fine-tuning the artifact-focused CNNs.
2. OpenForensics Dataset (Le et al., 2021):
 - A comprehensive 190,000-image dataset organized into three distinct subfolders, each containing unique real/fake western facial image pairs.

3.2.2 Justification for Selection:

The joint use of these two datasets is indispensable for the attainment of the research objectives. They were chosen because:

1. Component Coverage (Objective 1): They both contain deepfakes created by GAN model and diffusion models, consequently directly enable the study of component-specific artifacts for all the target architectures mentioned.
2. Benchmark Compatibility (Objective 3): Both datasets are considered a good representative standard deepfake detection evaluation sets and are used to measure final detection performance reliably against the existing methods.

3. Fidelity Testing (Scope): StyleGan2, being a dataset of high-fidelity fakes, is instrumental in ensure that the algorithm is challenging with hyper realistic deepfakes that look too real.

3.2.3 Data Preprocessing

In order to provide uniform input for artifact extraction and model training, data preprocessing steps are performed by utilize the OpenCV and PyTorch libraries from Python. Here is the following preparations done on the data:

1. Standardized Resolution: All images are resized to 256x256 pixels to maintain consistency across the five distinct model architectures.
2. Normalization:
 - ResNet-50, Xception, Vision Tranformer (sigmoid, 3 channels): Normalized using ImageNet stats (mean=[0.5, 0.5, 0.5], std=[0.5, 0.5, 0.5]) to align with pretrained model expectations.
 - DFT/Canny images (sigmoid, 1-channel): Min-max normalized to [0, 1] to reduce computational complexity.
3. Labeling & Splitting:
 - Assign digits 0 and 1 to real and fake images correspondingly. Besides, generation models used as the sub-labels to help in tracing of artifact-source relationships.

3.3 Component-Specific Artifact Extraction

The scope of work has limited to finding the artifacts that appear only in specific datasets used in this experiment. All the operations are carried out in Python with OpenCV for edge detection and PyTorch for frequency analysis.

3.3.1 DFT for High-Frequency Artifact Extraction

Most of the generative models had sacrificed high-frequency details due to latent regularization. Diffusion models introduce "frequency bias" such as high

frequency components who suppressed when denoising process. Discrete Fourier Transform (DFT) changes the images to frequency domain where these artificial can be measured.

Implementation Steps:

1. **DFT Conversion:** Use `cv2.dft()` to preprocessed grayscale facial image, then convert pixel values in spatial domain to frequency coefficients. Center the frequency spectrum with low frequencies and high frequencies at the edges using `np.fft.fftshift()`.
2. **High-Frequency Masking:** Create a circular mask (radius = 50 pixels) to cut off high-frequency components. This can targeting the frequency range where the artifacts of diffusion model are most visible (Wang et al., 2021).
3. **Artifact Quantification:** Measure the High-Frequency Energy (HFE) of the masked spectrum:

$$HFE = \sum_{u,v \in mask} |F(u,v)|^2$$

Where the DFT coefficient at frequency, lower HFE indicates more severe high-frequency loss (a key indicator of VAE/KSampler fakes).

3.3.2 Canny Edge Detection for Boundary Artifact Extraction

U-Nets can cause "edge blending artifacts" in uneven transitions between facial regions and backgrounds due to skip-connection feature merging. Canny Edge Detection highlights edges, making these blending inconsistencies visible.

Implementation Steps:

1. **Edge Detection:** Apply `cv2.Canny()` to preprocessed images with threshold values $T1 = 50$ and $T2 = 150$ (optimized for facial edges via validation set testing). This generates a binary edge map (1 = edge, 0 = non-edge).
2. **Boundary Artifact Identification:** Use `dlib`'s facial landmark detector to define a facial boundary, key points like jawline, eyebrow edges. Focus on edges along this boundary, where U-Net blending artifacts occur.

3. **Artifact Quantification:** Compute the **Edge Irregularity Score (EIS)** for boundary edges:

$$EIS = \frac{\text{Number of disconnected edge segments}}{\text{Total length of boundary edges (pixels)}}$$

Higher EIS indicates more severe blending artifacts (a key indicator of U-Net fakes).

3.4 Detection Algorithm Development

The core of the detector system is a multi-model ensemble that combines extracted artifact features (HFE, EIS) and RGB features from state-of-the-art architectures. It is built in Python using PyTorch for model training and scikit-learn for feature processing.

3.4.1 Feature Fusion

The ensemble integrates five models—three RGB-based (ResNet50, Xception, ViT) and two artifact-focused (DFT CNN, Canny CNN)—with voting-based fusion. The architecture is visualized in Figure 2.

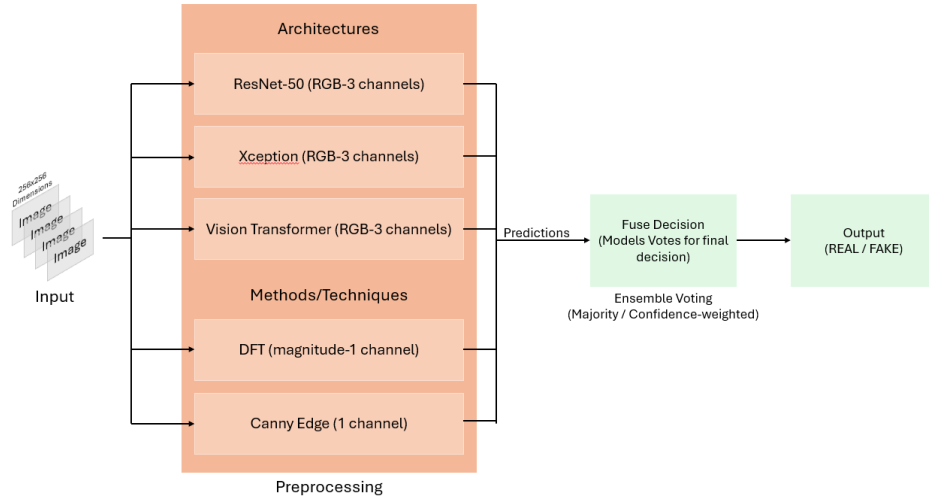


Figure 3.1: Ensemble Model Architecture and Detection Flow

The ensemble processes 256x256 preprocessed images through five parallel models. RGB models (ResNet50, Xception, ViT) extract high-level visual features, while DFT/Canny CNNs target component-specific artifacts. Predictions are

aggregated via majority voting to produce the final "REAL/FAKE" output, with forensic evidence (DFT magnitude/Canny edges) for interpretability.

3.4.2 Feature Fusion & Voting Strategy

Feature fusion and prediction aggregation are designed for interpretability and generalization:

1. **Feature Fusion:** Each model processes its input (RGB/DFT/Canny) independently to generate a "REAL/FAKE" prediction with confidence (0–100%).
2. **Voting-Based Aggregation:** The `ticket_based_voting` function aggregates predictions:
 - Each model gets 1 vote (majority wins).
 - Ties are resolved by averaging confidence scores (e.g., if 2 models predict "REAL" and 2 predict "FAKE," the side with higher average confidence determines the outcome).
 - Rationale: Fusion mitigates individual model weaknesses (e.g., RGB models fail on unseen generators; forensic models miss subtle fakes).

3.4.3 Training Parameters

- **Optimizer:** apply Adam with learning rate = 0.001 and weight decay $\leq 1e-5$ to balance convergence speed and overfitting.
- **Loss Function:** Binary Cross Entropy.
- **Training Epochs:** 8~10
- **Platform used:** Google colab

3.5 Interactive Detection Interface (Flask App)

A Flask web interface is proposed and will be implement to make the ensemble accessible for practical use. Key features will include:

1. **Input Support:** Single images (JPG/PNG/BMP), multi-image batches, or ZIP files (batch processing).

2. **Model Management:** Load/save models (supports .pt, .pth, .safetensors); auto-detect model type (RGB/DFT/Canny); switch between CPU/GPU.
3. **Output Visualization:**
 - Per-image prediction (REAL/FAKE) + confidence score.
 - Per-model results (for ensemble interpretability).
 - Forensic evidence: DFT magnitude spectra, Canny edge maps.
 - Batch summary stats: Total images, real/fake/error counts, average confidence.

Performance plots: Confusion matrices, confidence distributions, result counts.

3.6 Experiment & Evaluation

3.6.1 Evaluation Metrics

Ensemble effectiveness is measured using four automated metrics, which calculated via scikit-learn:

1. **Accuracy:** Overall correctness ($\frac{TP+TN}{TP+TN+FP+FN}$).
2. **Precision:** Minimize false positives ($\frac{TP}{TP+FP}$), for avoiding false accusations of real images.
3. **Recall:** Minimize false negatives ($\frac{TP}{TP+FN}$), to detecting hidden fakes.
4. **F1-Score:** Harmonic means of precision and recall ($2 \times \frac{Precision \times Recall}{Precision + Recall}$), accounts for dataset imbalance.

3.6.2 Experiment Environment

- **Hardware spec:** Intel Core i7-13700HX CPU, NVIDIA RTX 4060 GPU (8GB VRAM).
- **Environment:** Python 3.10.11, OpenCV 4.9.0.80, PyTorch 2.5.1, Torchvision 0.20.1, numpy<2, Flask, scikit-learn, timm.

3.7 Summary

This methodology is interpretable, reproducible, and targeted to component-specific artifacts. By combining RGB architectures with DFT/Canny-based forensic analysis, the ensemble addresses the limitations of black-box models and surface-level cue methods. The Flask interface adds practical utility, while standardized datasets and metrics ensure results can be benchmarked against existing research, which directly indicate supporting the project’s goal of improving deepfake image detection technology.

CHAPTER 4

DATA ANALYSIS & RESULTS

4.1 Experimental Setup

The chapter evaluate the proposed hybrid ensemble deepfake detector and make a comparison with individual baseline models. Two Kaggle datasets were used for experiment: (i) a StyleGAN2-based dataset that consists of 13,000 images, and (ii) OpenForensics dataset that approximately 190,000 images. The model training was done in Google Colab, while the testing and evaluation were experiment through local pipeline implementation. Were exactly same preprocessing setting, 256×256 resizing and normalization was used for all models to be able to compare results fairly, and there is six detectors have been tested: ResNet50, Xception, Vision Transformer (ViT), a CNN based on DFT, a CNN based on Canny edge, and the proposed hybrid ensemble which is a majority voting model.

4.2 Evaluation Metrics and Interpretation

The performance of model is presented through accuracy, precision, recall, and F1 score metrics. In this research, precision/recall/F1 are figured with the 'FAKE' class as the positive class mainly because the cost of false negatives (fake predicted as real) is considered to be very different from the cost of false positives (real predicted as fake) in forensic and cybersecurity domains.

Accuracy simply calculates the total number of correct predictions over the two classes. Precision (FAKE) accounts for when the model falsely accuses real images. Recall (FAKE) counts deepfakes that the model failed to catch. F1 score is the harmonic mean of precision and recall and, unlike accuracy, it is less sensitive to class imbalance or distribution shift.

4.3 Results on StyleGAN2 13k Dataset

Model	Overall Accuracy (%)	Overall Precision (%)	Overall Recall (%)	Overall F1-Score (%)
Resnet	100.00	100.00	100.00	100.00
Xception	100.00	100.00	100.00	100.00
Vision Transformer	100.00	100.00	100.00	100.00
DFT	91.00	100.00	91.00	95.29
Canny edge	99.00	100.00	99.00	99.50
Proposed Ensemble	100.00	100.00	100.00	100.00

Table 4.1: Performance on StyleGan2 13k Dataset

Table 4.1 showed summary of the findings of StyleGAN2 dataset after testing 100 pieces deepfake images. All RGB-based, 3 channels models (ResNet50, Xception, and ViT) obtained perfect scores of 100% across four metrics, i.e., accuracy, precision, recall, and F1 score. The proposed ensemble also scored 100%. Canny model attained almost perfect results (99% accuracy, 99% recall, $F1 = 99.50\%$) while the DFT model did not perform very well (91% accuracy, 91% recall, $F1 = 95.29\%$) even though it is 100% precision.

The excellent good results here imply that StyleGAN2 dataset is very distinct for current experimental setup. Which means the models have found very clear and consistent features separating real images from fake ones. However these great results might also be due to the uniform properties of the dataset, such as similar generation methods or preprocessing styles. Hence, more evaluation results like confusion matrices and sample predictions can additionally demonstrate the strength and trustworthiness of the proposed system.

4.4 Results on OpenForensics Dataset

Model	Overall Accuracy (%)	Overall Precision (%)	Overall Recall (%)	Overall F1-Score (%)
Resnet	97.00	100.00	97.00	98.48
Xception	100.00	100.00	100.00	100.00
Vision Transformer	100.00	100.00	100.00	100.00
DFT	73.00	100.00	73.00	84.39
Canny edge	86.00	100.00	86.00	92.47
Proposed Ensemble	99.00	100.00	99.00	99.50

Table 4.2: Performance on OpenForensics Dataset

Table 4.2 shows the results of experiments conducted on OpenForensics dataset. The proposed ensemble method accuracy went up to 99% accuracy, 100% precision, 99% recall, and a F1 on 99.50%. Xception and ViT were showing 100% results across the board, whereas ResNet50 fell a little (97% accuracy, 97% recall, F1 = 98.48%). Artifact-only detectors demonstrated a bigger fall off, DFT model is plummeted to 73% accuracy and recall, with a F1 on 84.39%, while Canny model 86% on accuracy and recall, with a F1 on 92.47%, both still marking 100% precision.

The second dataset has higher visual and generative variability than StyleGAN2 dataset, which can makes classification task harder. Under these circumstances, detection methods that depend only on one type of feature, like frequency-based or edge-based cues, may show less stability and consistency. On the other hand, the suggested ensemble method still delivers trustworthy performance by combining different and complementary feature viewpoints. Such multi-cue fusion helps the system to be more flexible with changes in image features and resulting in improved generalization skill.

4.5 Comparative Analysis and Robustness

On both datasets, the showed RGB-based models have been always high-performing, whereas the artifact-based detectors have displayed lower and more fluctuating results when applied independently. This implies that single models vary in their trustworthiness across datasets.

The proposed ensemble demonstrates more stable behavior by combining the outputs of multiple detectors. When one model produces an incorrect or uncertain prediction, other models may carry the final decision for it. The use of majority voting with confidence-based tie-breaking reduces prediction variability, which is reflected in the ensemble’s strong recall on the larger dataset while maintaining perfect precision.

4.6 Confidence Behaviour and Error Characterisation

Confidence distributions play a crucial role in operational decisions as mistake made with a high level of confidence tend to be more worse than low confidence. In this scheme, every single model emits a confidence figure which is used to decide on voting ties in case the average confidence of the models has the smallest difference. This way of aggregating dramatically reduces the levels of uncertainty and at the same time fosters consistent judgments when several models are in agreement.

4.7 Summary of Findings

In essence, StyleGAN2 dataset enabled almost perfect performance for most of the models, whereas the OpenForensic dataset revealed robustness differences. Artifact-only detectors were vulnerable to dataset change, but the suggested hybrid ensemble consistently showed high performance (99% accuracy and 99% recall) and flawless precision on trained dataset, which more fault tolerance than a single-stand model. These findings provide evidence to support the concept that the fusion of multiple cues makes deepfake image detection not only more stable and robust but also more interpretable.

<i>Model</i>	<i>StyleGAN2 Dataset (%)</i>	<i>OpenForensic Dataset (%)</i>
<i>ResNet50</i>	99.96	99.38
<i>Xception</i>	100.00	99.44
<i>ViT</i>	99.54	99.67
<i>DFT-based</i>	58.45	71.06
<i>Canny-based</i>	97.14	95.10
<i>Proposed Ensemble</i>	91.79	95.13

Table 4.3: Average Confidence Scores on 100 Fake Images

The average confidence analysis shown are information about the level of certainty of each model when deepfake making predictions. Deep learning models such as ResNet50, Xception, and ViT kept confidence values at levels higher than 99% both on one dataset and another. Hence, the implication is that these three neural network architectures are extremely good on discovering subtle patterns that differentiate genuine image from fakes.

In contrast for DFT model was found to give considerably lower confidence scores. This indicate that the fact, frequency-domain features alone may be more sensitive to changes in the dataset characteristics and may not generalize as well across different manipulation styles. On the other hand, Canny model kept relatively high confidence, which means that edge level inconsistencies are still reliable indicators of image manipulation.

Notably, the proposed ensemble detector has rather moderate confidence values. This characteristic indicates that the decision making process is based on multiple features interpretation, thus the ensemble detector can deliver more reasonable, trustworthy, and stable performances in different datasets.

4.8 Discussion of Key Findings

The outcomes strongly support recurring idea in deepfake forensics field: comprehensive detection is achieve through multi-domain evidences. Deep convolutional networks trained in the RGB domain can deliver powerful performance, and provided test distribution that identical to the training set. However, these methods might be use on dataset specific with different artifacts. Forensic methodologies such as frequency statistics and analysis of the irregularities along the edge boundaries have a clear interpretation to forensic analyst, but they still cannot serve as universal discriminators on cross domain data. On the other hand, the hybrid ensemble can resolves this issue by fusing different complementary representations and thereby make it more stable and accurate.

Moreover to that, the architecture of the tool also enhances the comprehensibility by producing the forensic clues (DFT magnitude and Canny edge lines). This is a great advantage in cybersecurity scenarios where the human operators are required not only to make a decision but also to provide reasons for their decisions and to carry out further investigations into why the particular image has been flagged as suspicious.

CHAPTER 5

CONCLUSION

5.1 Conclusion

In this project tackled the issue of deepfake detection by building a hybrid ensemble-based system for detecting synthetic image. The main goal was to find out blending signals such as semantic RGB features, frequency-domain artifacts, and structural boundary inconsistencies that better accuracy, stable and generalization than using single-model methods.

Theoretically, RGB based deep learning models like ResNet50, Xception, and Vision Transformer can do very well on training data only as they have high representational power. On the flip side, their performance may become unpredictable when they face images generated by different or completely unknown generation patterns, so ensemble architecture is proposed in this report. On the other hand, artifact detectors utilizing features in the frequency domain or edges have their pros and cons. They are able to detect very subtle changes in images but it's not reliable in the future since deepfake keep evolving each years.

The hybrid ensemble model that was suggested combines these five types of detection through a voting-based fusion mechanism. The idea here is that the deficiencies of a certain model can be made up for by other models in such a way that overall stability and consistency of performance across different datasets can achieved. Experimental results confirm that mixing of different cues not only makes the system more robust and less sensitive to changes in data distribution but also more interpretable by revealing different forensic aspects.

This paper validates ensemble-based deepfake detection more efficient and accurate than one single detection, but it is not worth in real world deployment due to big training workload just to achieve slightly more accuracy. The system presented

here serves as a viable and scalable platform which future research on deepfake forensics can build on.

5.2 Limitations

This research, although showing promising results, has several limitations. To begin with, the system evaluation was only on image-based deepfake datasets and the question of whether it can be effective in video-based deepfakes remains untested. Secondly, the ensemble architecture increases the computational cost of the system over single-model detectors, which could lead to non-compliance issues with real-time or resource-constrained settings. Thirdly, the paper considers only frequency and structural artifacts detection in binary classification output, while this not included other forensic clues such as biological signals, illumination inconsistencies and temporal artifacts.

Moreover, the datasets deployed in this research are indeed large, but still might not capture the entire spectrum of real world deepfake generation techniques due to the variety of deepfake architecture builds. As generative models keep progressing, fresh manipulation patterns will likely come up that will be a challenge for existing detection methodologies.

5.3 Future Work

In the future work, this research can consider work along this line that investigates a totally disparate paradigm of deepfake detection. For example, a real centric learning framework might be created where the model is developed mainly on authentic images showing real intrinsic features in multiple feature domains, such as RGB semantics, frequency domain statistics, and structural boundary patterns.

This method allows for a redefinition of the framework as an anomaly aware or open set detection system, where the model is trained to identify real images with great certainty and to label unfamiliar or unknown patterns as "unknown." Thus, using a multi level confidence or tertiary decision framework, where the outputs denote real, uncertain, or anomalous samples, one can accomplish this. This approach makes it

possible to communicate uncertainty openly, which is particularly critical when deploying the system in the real world where there will be many manipulations that have not been seen before.

Moreover, one can collect images of real scenes that belong to various camera sources, brands, and imaging pipelines to make the real image corpus more comprehensive and thus the modelling process more realistic and robust. Today's smartphones implement different computational photography ideas and AI based enhancers that significantly affect the statistical features of real images. Equipped with the knowledge regarding the differences in real images taken by different devices, the detector will be able to differentiate between natural variations in images and artificially manipulated ones more accurately.

Instead of limiting the number of end classes to three, one may investigate a hierarchical or multi level real image distribution characterization. To this end, various latent layers correspond to the different real image domains, while the anomaly region is allocated to the out of distribution samples. Here the detector can not merely make judgments about the authenticity of an image, but also about the underlying reasons and modalities of its pattern divergence from the real world ones.

Integrating interactive artificial intelligence (XAI) techniques into the system is another avenue that such work could pursue. Such a system that is aware of the feature domains that contribute to uncertainty or anomaly detection could offer understandable explanations of the visual aspects that seem abnormal or distorted. Such interpretations will not only help the users to trust the system but will also assist in forensic analysis.

In short, a method that goes beyond a simple binary classification by employing real centric, a "tertiary system" where 0, 1 as real indication, and 2 label as unknown field, which is deepfake. Not signmoid, not softmax, but a more layer classification on different field of "real" media. Uncertainty aware detection techniques is a promising avenue of research that is more in line with the challenges faced by real world deployments of deepfake.

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APPENDICES

Appendix A: Meeting Logs



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 5 October 2025	MEETING NO.: 1
PROJECT ID: T884174	
PROJECT TITLE : Identifying Deepfake Image	
SESSION : 2025/2026	COMPANY SUPERVISOR :
STUDENT ID & Name:	FACULTY SUPERVISOR :
ONG YONG QUAN 243UT246XG	Ts. Dr. Ho Yean Li

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] • Research Foundation: Analyzed deepfake generation technologies (GANs, LDMs, VAEs, U-Nets, KSamplers) to ensure identification features match real-world operations. • Resource Preparation: Identified and sourced open-source datasets (FaceForensics++, Celeb-DF) and selected core libraries (PyTorch, OpenCV, dlib) for implementation. • Strategy Formulation: Proposed a component-targeting approach focusing on specific defects like VAE high-frequency blur and U-Net edge inconsistencies. • Drafting: Composed the project Abstract and Acknowledgements.
2. WORK TO BE DONE Conduct a literature review with at least 5 references to position the component-artifact approach.
3. PROBLEMS ENCOUNTERED Currently none.
4. COMMENTS


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Company Supervisor's
Signature & Stamp

Faculty Supervisor's
Signature & Stamp


Student's Signature



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 12 October 2025	MEETING NO.: 2
PROJECT ID: T884174	
PROJECT TITLE : Identifying Deepfake Image	
SESSION : 2025/2026	COMPANY SUPERVISOR :
STUDENT ID & Name:	FACULTY SUPERVISOR :
ONG YONG QUAN 243UT246XG	Ts. Dr. Ho Yean Li

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] - Chapter 1 (Introduction): Completed the Background, Problem Statement, Aim & Objectives, and Scope. - Chapter 2 (Literature Review): Documented the evolution of deepfakes, existing detection methods, their limitations, and the research gap.
2. WORK TO BE DONE Develop the methodology section, including system framework design and initial image preprocessing.
3. PROBLEMS ENCOUNTERED Need to ensuring the "component-level artifact" focus is clearly distinguished from generic detection methods in the literature review.
4. COMMENTS


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 Student's Signature



Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 21 October 2025	MEETING NO.: 3
PROJECT ID: T884174	
PROJECT TITLE : Identifying Deepfake Image	
SESSION : 2025/2026	COMPANY SUPERVISOR :
STUDENT ID & Name: ONG YONG QUAN 243UT246XG	FACULTY SUPERVISOR : Ts. Dr. Ho Yean Li

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.] • Almost Complete Methodology (~90% done), including dataset specs (FaceForensics++, Celeb-DF), artifact extraction (DFT for HFE, Canny for EIS), lightweight neural network, and evaluation metrics. • Report Refinement: Adjusted the Introduction to include 3 independent problem statements and refined the Literature Review with seminal papers and benchmark stats (e.g., XceptionNet performance) as per your feedback.
2. WORK TO BE DONE Plan to continue working on the Methodology section. And put more detailed statistics for literature review.
3. PROBLEMS ENCOUNTERED To balance the depth of statistical analysis, we need to provide a strong benchmark for the proposed model.
4. COMMENTS


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Student's Signature

MEETING DATE: 12 January 2026	MEETING NO.: 4
PROJECT ID: T884174	
PROJECT TITLE : Identifying Deepfake Image	
SESSION : 2025/2026	COMPANY SUPERVISOR :
STUDENT ID & Name:	FACULTY SUPERVISOR :
ONG YONG QUAN 243UT246XG	Ts. Dr. Ho Yean Li

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.]

Focused entirely on the technical implementation to ensure the code functions as the "backbone" of the project. I have synchronized the FYP report to match the following integrated features:

- **Ensemble Integration:** Updated the report to reflect the shift from a single model to a Multi-Model Ensemble (ResNet50, Xception, ViT). This ensures the methodology aligns with the "Voting-Based Fusion" logic now present in the code.
- **Forensic Feature Alignment:** Documented the integration of Discrete Fourier Transform (DFT) and Canny Edge Detection. The report now explains how these specifically target high-frequency artifacts (HFE) and edge inconsistencies (EIS) found in the script.
- **Web Dashboard & Evaluation:** Updated the System Architecture section to include the Flask web interface and the automated Scikit-Learn evaluation pipeline (Accuracy, Precision, Recall, F1) that I have fully coded.

2. WORK TO BE DONE

Refine and fix the code to stabilize accuracy and begin Chapter 4 (Results).

3. PROBLEMS ENCOUNTERED

The system is functional but requires debugging to fix fluctuating prediction confidence.

4. COMMENTS


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Faculty of Information Science and Technology (FIST)
Final Year Project Meeting Log

MEETING DATE: 17 January 2026	MEETING NO.: 5
PROJECT ID: T884174	
PROJECT TITLE : Identifying Deepfake Image	
SESSION : 2025/2026	COMPANY SUPERVISOR :
STUDENT ID & Name:	FACULTY SUPERVISOR :
ONG YONG QUAN 243UT246XG	Ts. Dr. Ho Yean Li

All to be filled in by student

1. WORK DONE [Please write the details of the work done after the last meeting.]

1. Retrain and refine my detector source code to stabilize the prediction accuracy and resolved the fluctuating confidence issues mentioned in the last meeting.
2. Implemented a new module that converts video files into frames, allowing detector to support .mp4 files for deepfake detection.
3. Currently, I'm halfway through updating the project report to align with the latest technical changes and the new video processing workflow. Added relevant citations claims throughout the report.

2. WORK TO BE DONE

Finish the remaining of the report. Codes is done.

3. PROBLEMS ENCOUNTERED

None.

4. COMMENTS


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Appendix B: LLC Attendance

FYP for FIST: Attendance List

Session 4: 12/12/2025

The following students have attended the FYP session (FIST) with the library.

Trainer:



MELISSA DE VALDA BINTI MOHD YATIM
Assistant Manager, Library
Information Skill & Research Support Unit
Multimedia University
Jalan Ayer Keroh Lama
75450 Melaka

Melissa De Valda Binti Mohd Yatim

Librarian, Tun Dr. Hasmah Mohd Ali Digital Library

55	Kelven Chew Khye Wee	1211111644	DURIAN FARM INFORMATION AND TRACEABILITY WEB APPLICATION
56	Dennis Wee Khor Siong	1211112008	Single-Sample Face Recognition for Biometric Security Applications
57	Muhammad Faris Aidel Bin Dahlan	1211112319	The Study of Bandwidth Recycling in 4G/5G
58	ONG YONG QUAN	243UT246XG	IDENTIFYING DEEPFAKE IMAGES
59	Michelle Theba A/P Jeffrey Paul Raj	1231302828	Dimension Reduction and Machine Learning for Multimodal Data
60	CHAI QI YANG	243UT246XE	LIVENESS DETECTION FOR MITIGATING DISPLAY REPLAY ATTACKS
61	Joshua Lai Jia Jian	1211102761	web-based wifi tracking system
62	LAU YONG PIN	243UT246XR	LANDMARK LOCALIZATION IN CNN-BASED MASKED FACE RECOGNITION
63	Junior Gregory Danker	1231302631	ENHANCING ENGAGEMENT IN ONLINE LEARNING THROUGH ADAPTIVE CONTENT DELIVERY
64	Chern Hao Wen	243UT246XN	DATA STORAGE SYSTEM BY PDF417
65	CYRUS RHYS RONNY	1231303085	Campus Network Monitoring and Intrusion Detection Simulation
66	Amira Tsara Bt Mohd Zamri	1231302647	Music Recommendation System Based on User's Facial Age and Gender Recognition
67	Bong Li Hen	243UT246WG	Lost and Found
68	LUKMANNULHAKIM BIN MUHAFF	1211109429	FOREST FIRE ANALYSIS USING GOOGLE EARTH ENGINE (GEE)

Appendix C: Checklist for Final Report Submission
Faculty of Information Science and Technology (FIST)

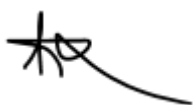
STUDENT'S DETAILS

Project Code	T88J474
Name	ONG YONG QUAN
ID No	243UT246XG
Title of Thesis	Identifying Deepfake Images
Supervisor Name	Ts. Dr. Ho Yean Li

REPORT ARRANGEMENT	√	Comments (if any differences)
1. Cover of The Final Report	√	
2. Title Page of the Final Report	√	
3 Copyright page of I Final Report	√	
4. Declaration Page of Final report	√	
5. Acknowledgement	√	
6. Table of Contents	√	
7. Abstract	√	
8. List of Tables	√	
9. List of Figures	√	
10. List of Symbols	√	
11. List of Appendices	√	
12. Chapter 1: Introduction	√	
13. Chapter 2: Literature Review	√	
14. Chapter 3: Methodology	√	
15. Chapter 4: Data Analysis & Results	√	
16. Chapter 5: Conclusion	√	
17. References – APA style	√	
18. Appendix A	√	
20. Appendix B	√	
21. Appendix C	√	

FORMAT OF REPORT	√	Comments
1. Page Numbering	√	
2. Font and Type Face	√	
3. Font Cover	√	
4. Tables and Figures	√	
5. Spine Format	√	
6. Comb Bind (For evaluation)		
7. Permanent Bind (After approval)		
8. Colour of the Front Cover		
9. Number of words > 10000 (Main content only)		

Checked by



Student's Signature & Date